MICHAEL MILLAN

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PROFESSIONAL SUMMARY

Mechanical Engineer (BS, UC Merced) with 2+ years of SolidWorks experience in CAD design, structural analysis, and data-driven design iteration. Experienced in full design-build-test cycles, FEA validation against real-world results, and parametric simulation to drive geometry decisions. Seeking a hardware development role where engineering rigor and iterative problem solving create measurable outcomes.

WORK EXPERIENCE

CAD Designer / Design Engineer | O&M Industries | Arcata, CA | *Feb 2025 – Present*

- Produce SolidWorks sheet metal and weldment models across light gauge to heavy structural plate in stainless steel, steel, aluminum, and exotic metals, generating complete drawing packages including flat patterns, BOMs, and fabrication notes for industrial clients including Tesla, Ceco Environmental, Adwest, and Nestec supporting large scale oxidizer systems, stainless ductwork assemblies, and custom process equipment fabrication.
- Use ProNest for flat pattern nesting and material optimization across high volume production runs and custom one-off projects; work directly in AutoCAD for 2D layout and field documentation.
- Manage drawing revision control across active projects, updating SolidWorks models and reissuing complete drawing packages to customers upon engineering change.

ENGINEERING PROJECTS

Kinematic Suspension Design/ Simulation System — *Personal Project — 2021 F-150* | *2025 – Present*

- Built full kinematic model in SolidWorks from 3D scan data, replicating OEM front suspension geometry including precise jounce/rebound travel, caster, camber, and toe characteristics.
- Executed parametric SolidWorks Design Studies iterating across 18+ configurations to converge on optimized geometry through data-driven comparison
- Built structured data analysis pipeline in Excel tracking four geometric parameters across full travel range, enabling quantitative design decisions across iterations
- Redesigned upper and lower control arm geometry within kinematic model to achieve long-travel packaging goals while preserving acceptable bumpsteer and camber curves.
- Project ongoing: currently transitioning from validated kinematic model to full 3D component design for fabrication.

Multi-System Mechanical Design/Fabrication Program — *Engineering/ Fabrication Project* | *2023 – Present*

- Established a full 3D scan-to-fabrication workflow using a Shining3D EinScan scanner — scanning complex vehicle geometry, cleaning mesh data, and importing into SolidWorks as reference surfaces for precise custom part design.
- Performed static structural FEA in ANSYS on fabricated aluminum assembly; simulation accurately predicted real-world failure location, validating the design-build-test methodology and informing corrective design iteration.
- Designed and fabricated a three-piece aluminum underbody skid plate system (front differential, transmission, transfer case)
- Designed and fabricated a roll-up aluminum truck bed cover, adapting structural reinforcements mid-build to accommodate material substitutions without compromising load integrity.
- Designed and fabricated a custom fairlead skid plate for a Badlands 12K winch using 3D scan data, sheet metal modeling, forming, and TIG welding.

EDUCATION

BS Mechanical Engineering | University of California, Merced | | *May 2024*

Welding Technology Certificate (09810.CN) | Merced Community College | | *Dec 2024*

TECHNICAL SKILLS

CAD / Design: SolidWorks (CSWA), AutoCAD, Fusion 360 sheet metal, weldments, assemblies, Design Study, simulation, kinematic modeling Fabrication & Tools: Shining3D EinScan 3D scanning, ProNest, TIG/MIG/SMAW welding, metal forming, plasma cutting, 3D printing Analysis: ANSYS FEA, SolidWorks Design Study, suspension geometry & kinematics, Excel data analysis and charting, iterative design and test cycles, root cause analysis